PREDICTING EMPLOYEE ATTRITION AT SAILFORT MOTORS

A Machine Learning approach for Identifying Attrition Drivers and Retention Strategies

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THE BUSINESS PROBLEM

The HR department at Salifort Motors wants to improve employee retention — but didn't know where to start.

With data collected from 15,000 employees, the goal was clear:

- What is making employees leave?
- Can we predict who is likely to leave — before they do?
- What can the company actually do about it?

Replacing a single employee costs an estimated 50–200% of their annual salary. At scale, attrition is not just an HR problem — it is a business risk.

REALITY CHECK

16.6%

Attrition Rate — 1 in 6 employees left

87%

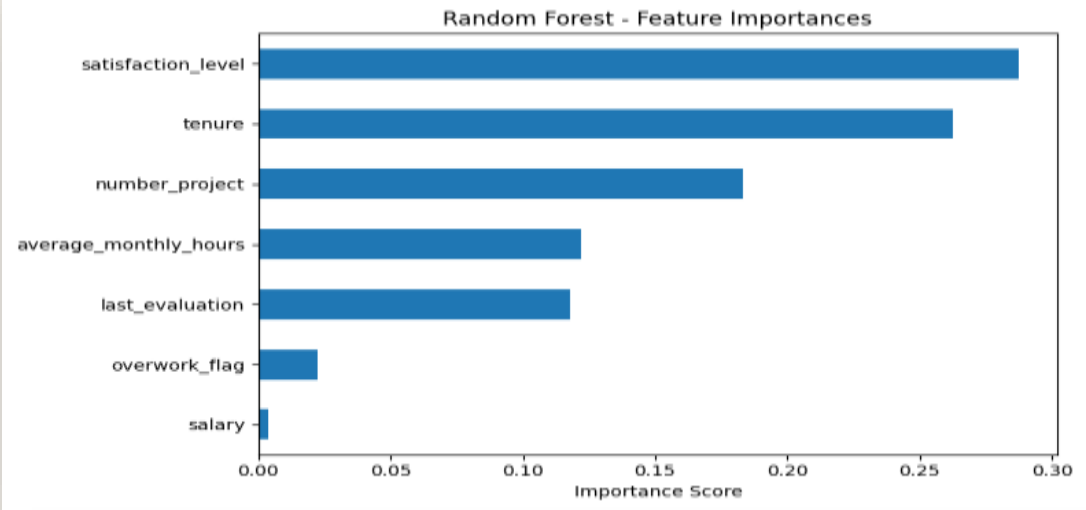
Of overworked employees left — 270+ hrs/month AND 6+ projects

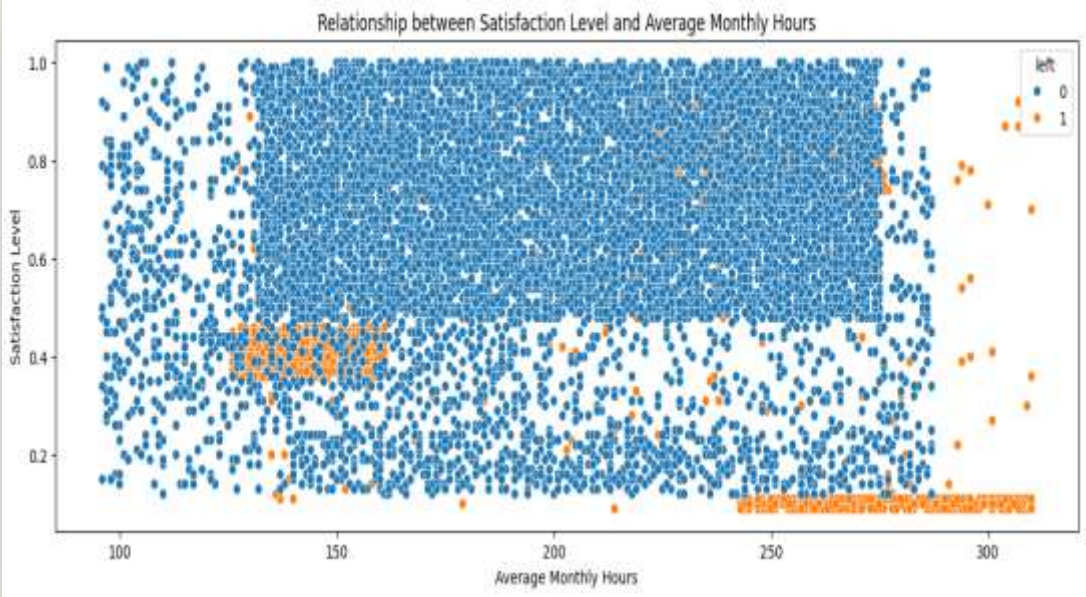
3–5×

Higher attrition among the lowest-satisfaction employees vs the average

This is not a uniform problem. It is concentrated, predictable, and preventable.

ATTRITION DRIVERS AND "BURNOUT" THRESHOLDS





Attrition is driven by a combination of **disengagement**, **workload pressure**, and **mid-career stagnation**.

- ***Satisfaction Level*** – The **strongest** and **most consistent** predictor. Lower satisfaction levels are directly associated with higher attrition, making **disengagement** the earliest warning signal.

- ***Workload*** – Employees working longer hours (**270+ hours per month**) or handling extreme project loads (**6+**) are significantly more likely to leave, with **7+** project holders leaving at **100% rate**, indicating sustained overwork.

- ***Tenure (3–5 Years)*** – Attrition peaks in **the mid-tenure window**, where employees reassess growth, compensation, and workload expectations.

- ***Performance Paradox*** – **High-performing employees** are leaving at **similar rates**, suggesting **burnout** rather than underperformance as the key driver.

Salary has a secondary influence, while factors such as department, promotions, and work accidents show minimal impact on attrition.

PREDICTIVE MODELLING

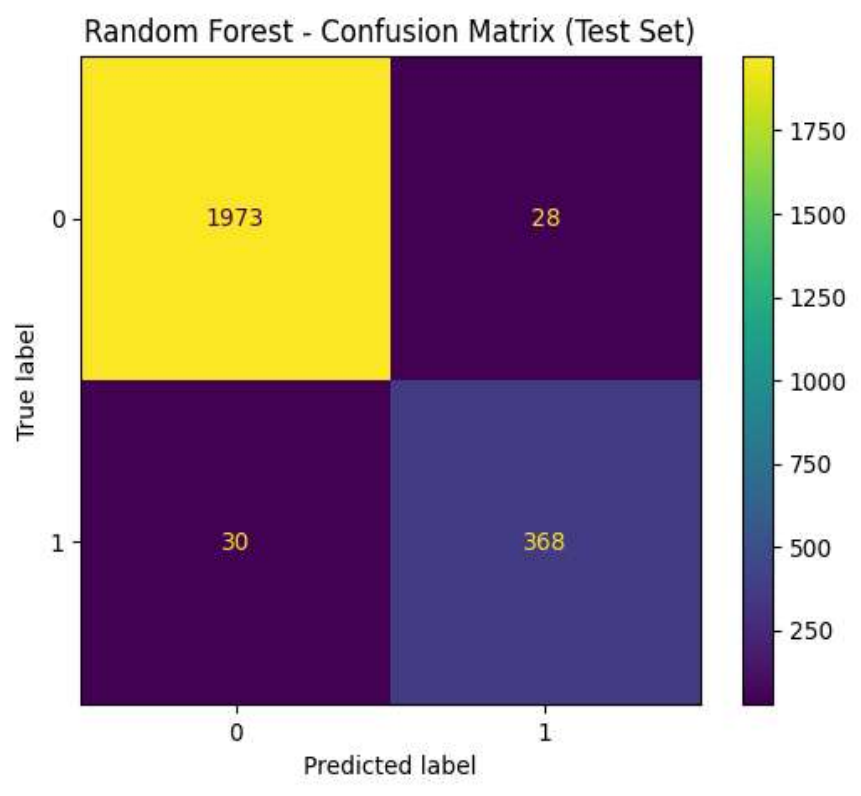
CHAMPION MODEL : RANDOM FOREST

The model can reliably identify employees at risk of leaving before attrition occurs.

- **92% Recall** → The model correctly detects 9 out of 10 employees who actually leave, ensuring most attrition cases are identified early.
- **93% Precision** → When the model flags an employee as "at risk," it is correct most of the time — meaning very few employees are incorrectly flagged.
- **97% Accuracy** → Overall, the model makes correct predictions in nearly all cases, making it reliable enough for real-world HR decisions.

Key Insight —The model identifies **high-risk employees early** with **minimal false alarms**, making targeted intervention both **practical** and **scalable**.

Model	Precision	Recall	F1	Accuracy
Decision Tree	0.33	0.94	0.49	0.67
Random Forest	0.93	0.92	0.92	0.97
XGBoost	0.70	0.94	0.80	0.92



STRATEGIC RECOMMENDATIONS

1. Enforce a hard ceiling on workload

Problem Identified:

Employees carrying 6 or more projects alongside 270+ monthly hours left at an 87% rate. This is not a risk factor – it is a near-deterministic outcome. The company is systematically overloading a subset of its workforce and losing them as a direct consequence.

Solution:

- Cap individual project assignments at 5. Any request to assign a 6th project should require explicit managerial sign-off with a documented justification.
- Flag any employee consistently exceeding 270 monthly hours for an automatic workload review the following month.
- Audit current project distribution across teams to identify whether overloading is concentrated in specific departments or roles – the EDA showed this is not random.
- Explore where automation or additional hiring can absorb excess workload rather than stacking it onto existing employees.

2. BUILD A RETENTION STRATEGY AROUND THE 3-6 YEAR TENURE WINDOW

Problem Identified:

Attrition peaks sharply in the mid-tenure range. These are fully trained, fully productive employees — the most expensive cohort to lose and replace. The data suggests Salifort is effective at onboarding but fails to sustain engagement past the initial phase, when employees begin benchmarking their growth and workload against their expectations.

Solution:

- Introduce structured career development conversations specifically at the 2-year and 4-year marks — before attrition risk peaks, not after.

- Create visible internal mobility pathways so mid-tenure employees see a future within the company rather than outside it.

- Assign a dedicated HR touchpoint for employees in the 3-6 year bracket whose satisfaction scores have dropped, prioritizing them for stay interviews before they reach the decision point.

- Review whether promotion and recognition cycles are actually reaching this cohort or whether they are being passed over in favor of newer or more senior employees.

3. REPLACE ANNUAL SATISFACTION SURVEYS WITH QUARTERLY PULSE CHECKS

Problem Identified:

Satisfaction level is the single strongest predictor of attrition in the entire model. Yet measuring it once a year means HR is always working with stale data. By the time a drop in satisfaction appears on an annual review, the employee has in most cases already made their decision.

Solution:

- Implement lightweight quarterly pulse surveys — 5 to 8 questions maximum — focused on workload, recognition, and growth. The goal is lead time, not comprehensiveness.

- Use the champion model each quarter by feeding updated satisfaction scores to generate a ranked list of at-risk employees. HR can then prioritize outreach based on model output rather than gut feel.

- Track satisfaction trends at the team level, not just individually. A team-wide drop is a management problem, not an individual one, and should trigger a separate intervention at the managerial level.

4. STOP TREATING HIGH PERFORMERS AS INFINITELY RESILIENT

Problem Identified:

Employees with high last_evaluation scores are leaving at rates comparable to low performers. The company appears to be rewarding strong performance with more work rather than with relief, recognition, or advancement — effectively penalizing its best people for being good at their jobs.

Solution:

- Decouple high performance from automatic workload increases. A strong evaluation score should trigger a conversation about growth and recognition — not an additional project assignment.

- Introduce performance-linked benefits that go beyond salary, such as flexible scheduling, project choice, or reduced mandatory hours — given that salary is the weakest predictor, non-monetary recognition is likely to have a stronger retention impact.

- Monitor the satisfaction-to-evaluation ratio as an internal metric. A high evaluation paired with a declining satisfaction score is the clearest early warning signal the data provides.

5. DEPLOY THE MODEL AS A LIVING HR TOOL

Problem Identified:

The model was built on historical data and evaluated at a single point in time. Attrition risk is not static — it evolves as employees change roles, accumulate projects, and shift in satisfaction. Treating this as a one-time analysis wastes its full potential.

Solution:

- Integrate the Random Forest model into a quarterly HR workflow where updated employee data is scored and an attrition risk ranking is produced for the HR team to act on.

- Define clear escalation thresholds — for example, any employee scoring above a defined risk probability triggers a mandatory check-in within 30 days.

- Retrain the model annually on fresh data to ensure it reflects any structural changes in the workforce, particularly if the company grows significantly or undergoes organizational restructuring.

- Use the model's output not just reactively but strategically — departments with consistently high predicted attrition should prompt a review of management practices, not just individual employee interventions.